Differential Expression and Co-expression Analysis of Tumour and Normal Tissue

Coursework Report for Essentials of Mathematics and Statistics

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1 Introduction

High-dimensional RNA-seq data are inherently stochastic and often characterized by over-dispersion, where the variance exceeds the mean. The primary objective of this report is to identify specific gene expression biomarkers that significantly distinguish between tumour and normal tissue. To ensure the statistical validity of these findings, the analysis employs probabilistic modelling to control for confounding factors, specifically sequencing depth and inter-patient biological variability.

2 Part 1: Data Exploration and Normalization

2.1 Data Structure and Characteristics

The dataset consists of two primary objects: an expression matrix `Y` and a metadata frame `patient_data`. The matrix `Y` contains raw gene counts for $4,568$ genes across 30 tissue samples, representing paired normal-tumour tissues from 15 patients. The `patient_data` object maps each column in `Y` to its clinical covariates: tissue type (Tumour/Normal) and patient identifier.

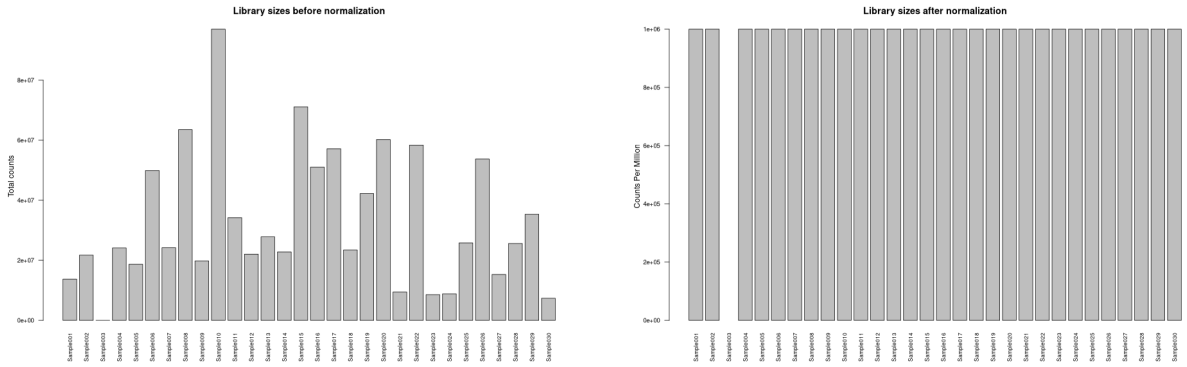
2.2 Library Size and Normalization

In RNA-sequencing, the "library size" (or sequencing depth) refers to the total number of reads generated for a specific sample. Variation in library size often arises from technical inconsistencies in the sequencing machine or library preparation rather than biological differences.

Preliminary analysis of the raw data revealed significant heterogeneity in sequencing depth (Figure 1a). Library sizes ranged from a minimum of 0 to a maximum of approximately 97 million, with a mean of 33.1 million (SD = 22.6 million). To account for this technical bias and render samples comparable, we applied a Counts Per Million (CPM) normalization:

$$\text{CPM}_{ij} = \left(\frac{\text{Raw Count}_{ij}}{\text{Library Size}_j} \right) \times 10^6 \tag{1}$$

Where i represents the gene and j represents the sample. CPM converts raw absolute counts into proportional abundance (Statomics n.d.).



(a) Library sizes before CPM normalization. (b) Library sizes after CPM normalization.

Figure 1: Library sizes before and after CPM normalization.

As illustrated in the post-normalization bar plots (refer to Figure 1b), the total normalized counts stabilize at one million for all viable samples, converting raw absolute counts into proportional abundance.

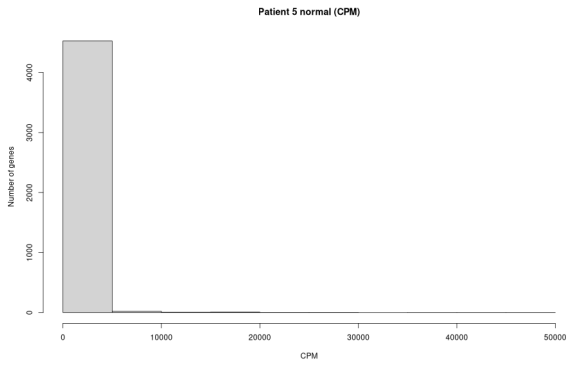
2.3 Distributions and Data Transformation

Raw RNA-seq count data is typically characterized by distinct skewness. Using Patient 5 (P5) as a representative case study, we examined the distribution of gene expression. The P5 tumour sample displayed a range of 0 to 16,713 CPM. The associated histogram (Figure 2c) revealed a severe right-skew: the vast majority of genes exhibited near-zero counts (sparsity), while a small subset of outliers exhibited extremely high expression.

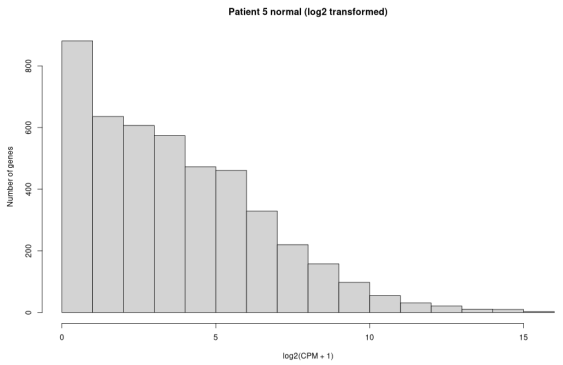
To prepare the data for downstream statistical testing, a $\log_2(\text{CPM}+1)$ transformation was applied. This transformation serves two purposes:

1. **Variance Stabilization:** It compresses the dynamic range, bringing highly expressed outliers onto a comparable scale with lower-abundance genes.

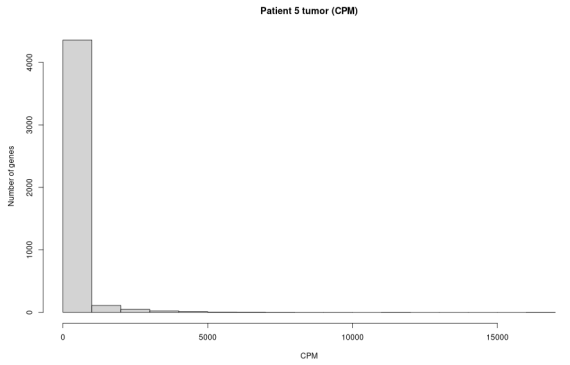
2. **Symmetry:** It transforms the skewed distribution into an approximate bell curve (Gaussian-like), which facilitates visualization and usage in linear models.



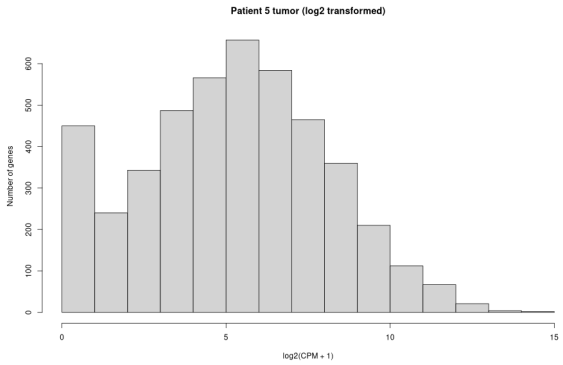
(a) Normal sample gene expression.



(b) Log-transformed normal sample gene expression.



(c) Tumour sample gene expression.



(d) Log-transformed tumour sample gene expression.

Figure 2: Patient 5 normal (top row) and tumour (bottom row) tissue before and after log transformation.

Quality Control and Sample Exclusion

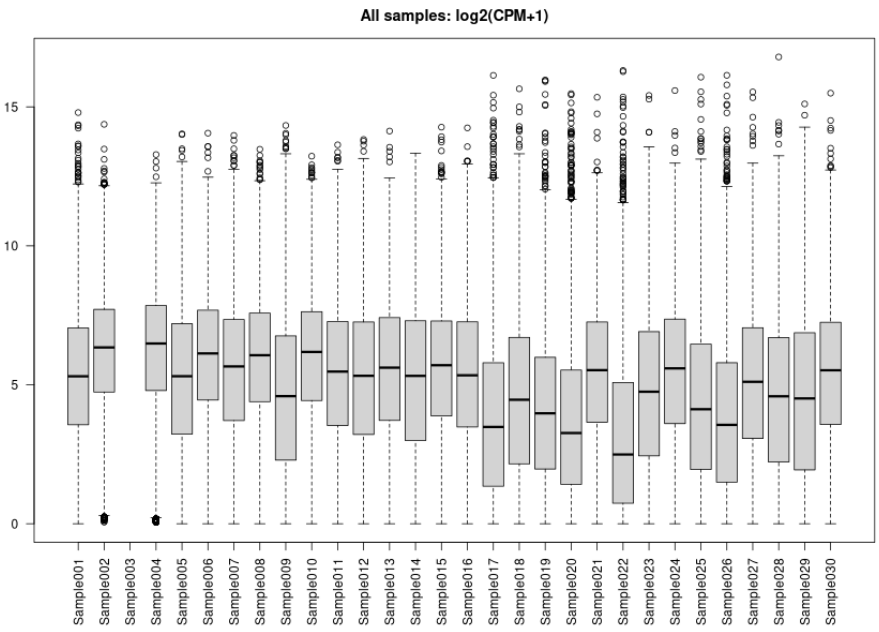


Figure 3: Log-transformed gene expression across all samples.

During the exploration of library sizes and distributions, `Sample003` was identified as an outlier with a library size of 0. This likely represents a technical failure during RNA extraction or data file corruption. As this sample contains no biological information, its inclusion would introduce mathematical errors (e.g., division by zero during normalization) and bias mean expression estimates. Consequently, `Sample003` was removed. To maintain the integrity of the paired study design, the corresponding matched pair (Patient 3 Normal; `Sample018`) was also excluded, leaving 28 samples for the final analysis.

2.4 Principal Component Analysis (PCA)

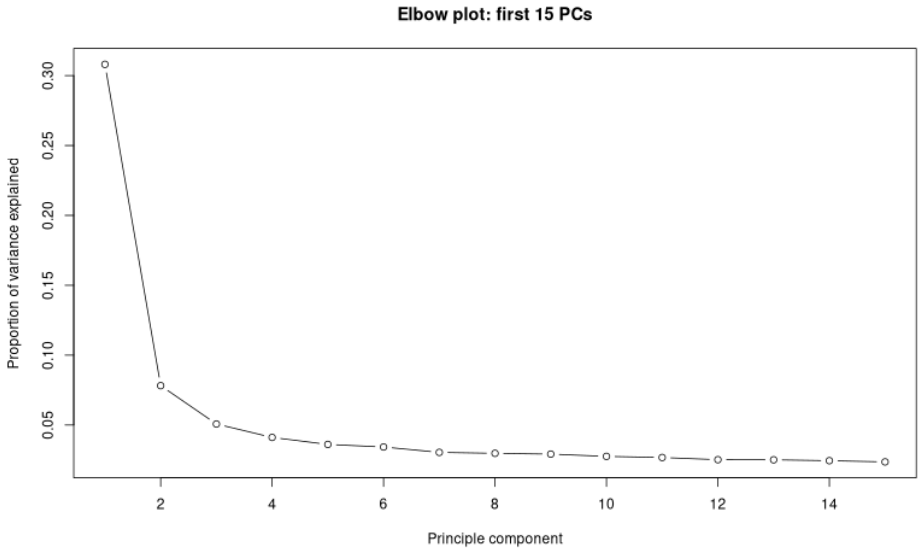


Figure 4: First fifteen principal components in PCA.

To assess the global structure of the data and identify major sources of variance, Principal Component Analysis (PCA) was performed on the cleaned, log-transformed dataset. The scree plot (Figure 4) indicates that the first two principal components (PC1 and PC2) capture the majority of the variance (30.8% and 7.8%, respectively) within the 4,568-dimensional gene space.

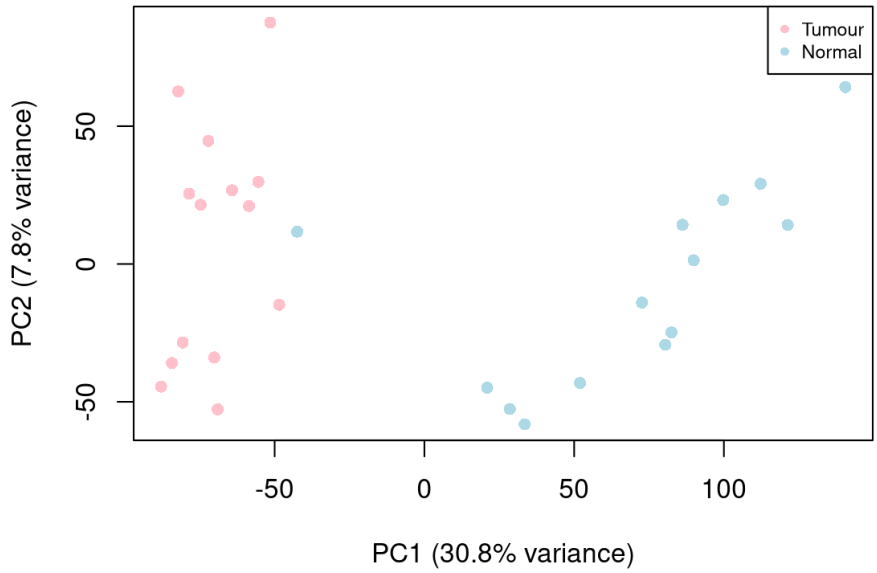


Figure 5: PC1 vs PC2.

Visualizing the samples on the PC1-PC2 plane (Figure 5) reveals distinct clustering patterns:

- **PC1 (Tissue Effect):** There is a clear horizontal separation between Tumour (Pink) and Normal (Blue) samples. This suggests that the strongest signal in the dataset is the biological difference between cancer and healthy tissue. Analysis of the loadings identified `ENSG00000104884` as a top contributor (loading = 0.0583) to this separation.

- **PC2 (Patient Effect):** The spread of samples along the vertical axis represents inter-patient variability (Tumour Heterogeneity). The fact that tumour samples do not cluster into a single tight point implies that while patients share a common cancer signature, individual genetics significantly influence gene expression profiles.

  To assess whether principal components reflect technical variation, we tested the correlation between PC1 scores and library sizes. A strong negative correlation was observed (Pearson $r = -0.85$, $p = 7.5 \times 10^{-9}$), indicating that sequencing depth substantially influences the primary axis of variation. This suggests CPM normalization incompletely removes library size effects, and results should be interpreted with this caveat.

3 Part 2: Differential Expression Analysis

3.1 Probabilistic Modelling Strategy

To quantify differential expression between tumour and normal tissues, we employed a generalized linear model (GLM) framework. Unlike continuous data, RNA-seq data consists of non-negative integers (counts) that exhibit **over-dispersion**, where the variance exceeds the mean (Var $> \mu$). This biological variability, driven by inter-patient heterogeneity, violates the assumptions of a standard Poisson model (where Var $= \mu$).

Consequently, a **Negative Binomial (NB)** regression model was selected. The NB distribution incorporates a dispersion parameter θ to model the mean-variance relationship as $\text{Var}(Y) = \mu + \theta\mu^2$, offering a robust fit for biological count data. Thus variance is modelled as a function of the mean and a gene-specific dispersion parameter rather than being fixed by the mean alone.

Model Specification

For each gene, the relationship between the expected count μ_i for sample i and the clinical covariates was modeled as:

$$\log(\mu_i) = \beta_0 + \beta_1 x_i + \text{offset}(\log(L_i)) \tag{2}$$

Where:

- x_i is the tissue type covariate (0 for Normal, 1 for Tumour).
- β_1 represents the log-fold change (the effect size of the cancer). The Null Hypothesis ($H_0 : \beta_1 = 0$) posits no differential expression.
- L_i is the library size. It is included as an **offset** (coefficient fixed at 1) rather than a variable, effectively modeling the *rate* of expression per read rather than the raw count.

The model was implemented in R using the `MASS` package:

```
glm.nb(y ~ x + lib_size)
# Note: lib_size enters as a predictor but acts as an offset
```

due to the log-link nature of the GLM.

3.2 Model Diagnostics and Fitting Issues

During the genome-wide application of the NB model, convergence warnings were generated for 492 out of 4,568 genes (approx. 10.8%).

These warnings primarily stem from **data sparsity**. Genes with high proportions of zero counts (dropouts) or insufficient variation between conditions provide limited information, making the Maximum Likelihood Estimation (MLE) numerically unstable. While some of these genes displayed statistically significant raw p-values, the instability of the model fit suggests these results may be unreliable (false positives). However, rather than discarding them immediately, they were flagged for cautious interpretation, as they may represent biologically relevant "on/off" expression patterns that strictly violate NB assumptions.

3.3 Statistical Significance and False Discovery Control

Testing thousands of genes simultaneously introduces a substantial Multiple Testing Problem. With a standard significance threshold of $\alpha = 0.05$, we would expect approximately 228 false positives purely by chance (0.05×4568).

To control the Family-Wise Error Rate (FWER), a Bonferroni correction was applied. FWER is the probability of making at least one false positive call in the set of 4,568 hypothesis tests, and Bonferroni guarantees this false positive at the chosen α (Clyte.tech 2023). This rigorous adjustment reduced the number of significant genes from 3,266 (raw) to 1,604 (adjusted). As illustrated in Figure 6, this filtering ensures that the reported biomarkers represent high-confidence signals rather than statistical noise. The diagnostic plot shows that while "warning" genes (red points) are scattered throughout, several survived correction, indicating strong signal magnitude despite fitting difficulties.

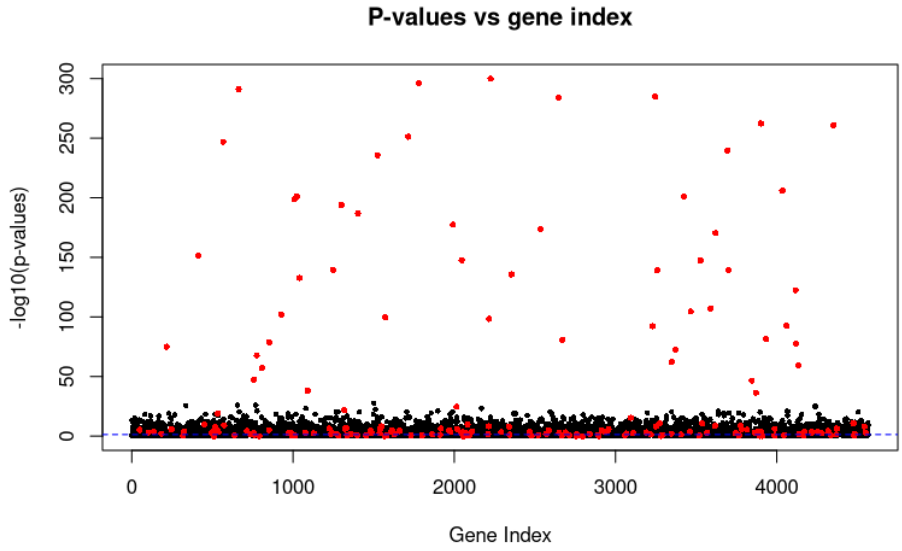


Figure 6: Raw p-values across genes for the tissue effect.

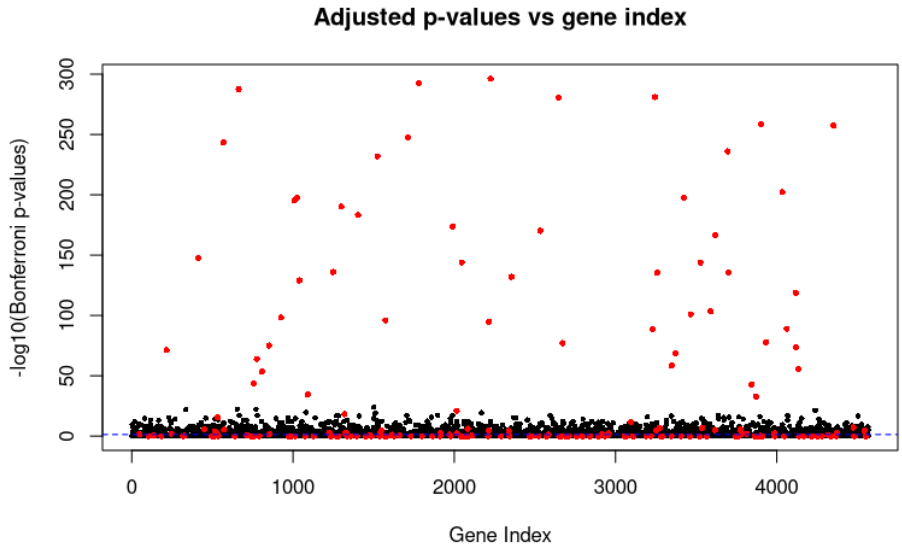


Figure 7: Bonferroni-adjusted p-values across genes for the tissue effect.

3.4 Alternative Approaches: Paired vs. Unpaired Linear Models

To validate our findings and assess the impact of the study design, we performed a sensitivity analysis using Linear Models (LM) on $\log_2$-transformed data. We compared two specifications:

1. **Model 1 (Unpaired):** Expression $\sim$ Tissue
2. **Model 2 (Paired):** Expression $\sim$ Tissue + PatientID

The Cost of Paired Analysis in Small Samples

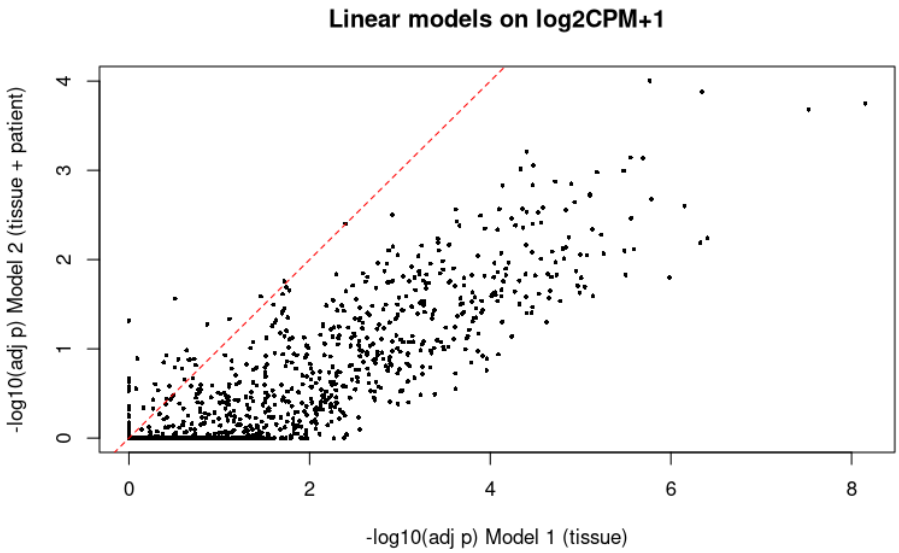


Figure 8: Comparison of -log10 adjusted p-values from Model 1 and Model 2.

A comparison of p-values (Figure 8) demonstrates that most points lie below the y=x line, indicating that Model 2 yields higher p-values (lower significance) than Model 1.

Clinical Interpretation: Contrary to theoretical expectations, the paired model (Model 2) yielded less significant results than the unpaired model (Model 1). This counterintuitive finding reflects a fundamental bias-variance trade-off in small samples. While including `PatientID` theoretically controls for inter-patient variability by allowing each patient to serve as their own baseline, it requires estimating 13 additional parameters (one per patient minus the reference level). With only 28 observations, this substantially reduces residual degrees of freedom (from 26 to 13), inflating standard errors and reducing statistical power.

In larger cohorts, the variance reduction from blocking on patient identity would be expected to outweigh this degrees-of-freedom penalty, and the paired design would likely demonstrate superior performance. This result underscores the importance of adequate sample size when employing complex study designs—the theoretical advantages of paired analysis can only be realised when sufficient data are available to estimate all model parameters precisely.

Model Comparison: NB vs. Linear

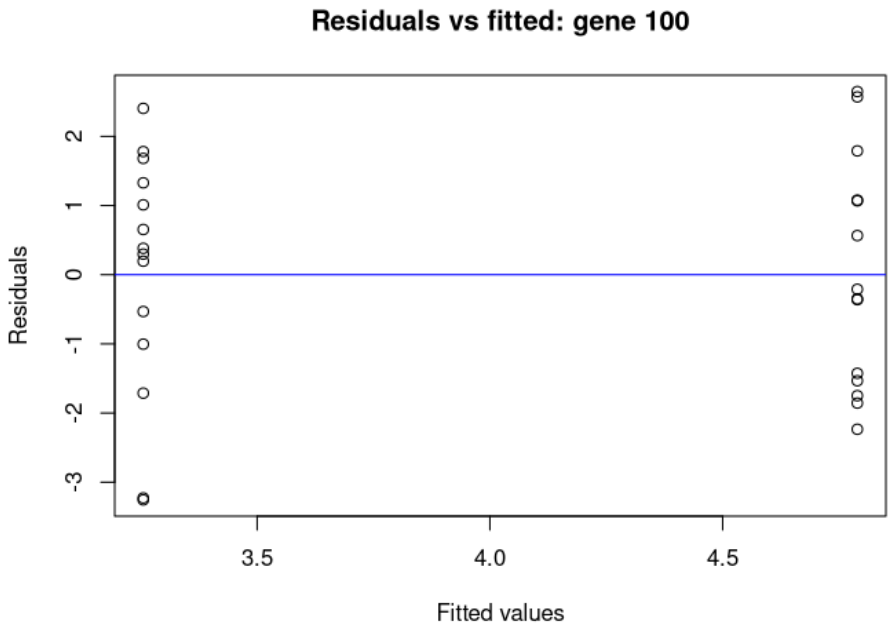


Figure 9: Residuals vs Fitted for Gene 100.

While the Linear Model on log-transformed data performed adequately (residuals showed approximate homoscedasticity, satisfying assumptions), the Negative Binomial model remains theoretically superior. The log-transformation distorts the error structure of discrete counts, whereas the NB model preserves the original data structure and explicitly models the over-dispersion inherent to RNA-sequencing.

4 Part 3: Exploratory Co-expression Analysis

Beyond identifying individual biomarkers, we investigated how gene interactions change between disease states. This "network-level" view can reveal regulatory pathways (e.g., cell cycle or immune response) that are disrupted in cancer.

4.1 Differential Co-expression (Network Rewiring)

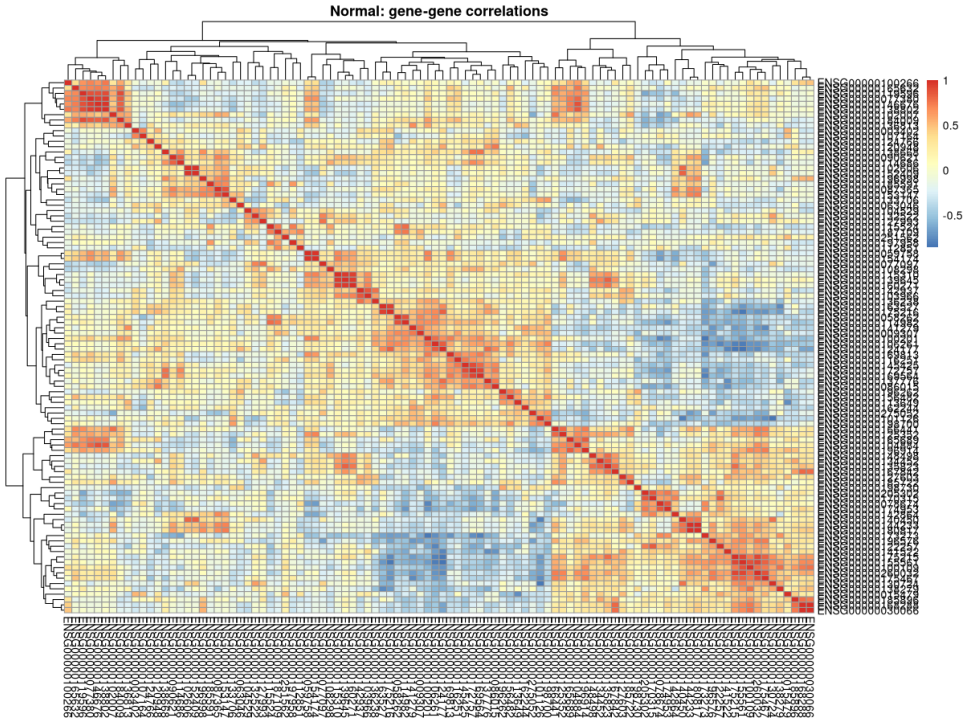


Figure 10: Gene-gene Pearson correlation heatmap for normal samples (top 100 genes).

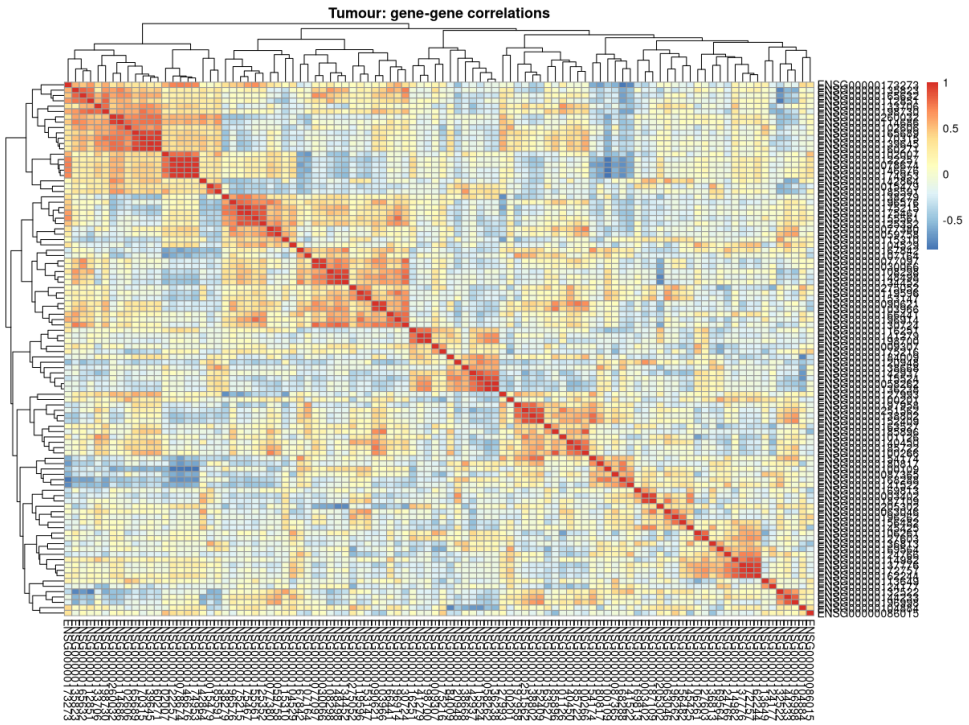


Figure 11: Gene-gene Pearson correlation heatmap for tumour samples (top 100 genes).

We first computed gene-gene Pearson correlations separately for Normal (Figure 10) and Tumour (Figure 11) samples using the top 100 most variable genes.

Interpretation:

- **Baseline Networks:** The Normal heatmap represents physiological homeostasis—the static regulatory relationships required for healthy tissue function.
- **Rewiring:** Differences observed in the Tumour heatmap suggest "network rewiring." A loss of correlation in the tumour samples implies that the tight regulation of specific pathways has broken down (dysregulation). Conversely, the emergence of new high-correlation blocks suggests the activation of tumour-specific regulatory programs, such as angiogenesis or proliferation signaling.

4.2 Correlation of Differences (Dynamic Co-regulation)

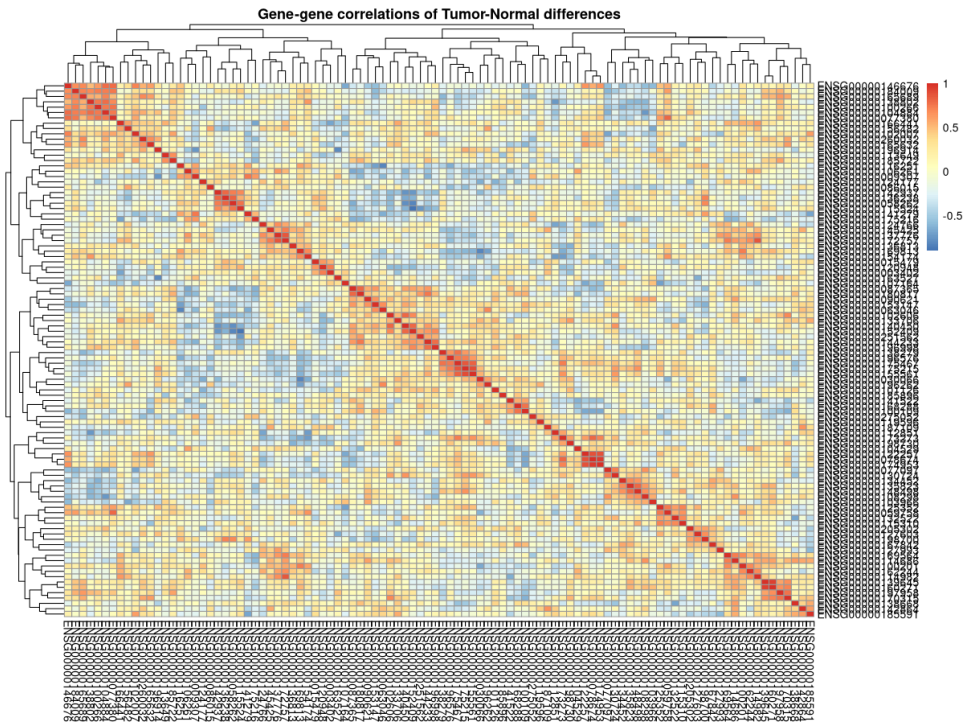


Figure 12: Gene-gene Pearson correlation heatmap for tumour-normal differences (top 100 genes).

Next, we calculated the *difference* in expression (Tumour minus Normal) for each patient and computed the correlation matrix of these differences (Figure 12).

Interpretation: This approach isolates the *dynamic change* caused by the disease. A high positive correlation between Gene A and Gene B in this plot indicates that

when Gene A increases in a tumour, Gene B also tends to increase in that same patient. This points to **co-regulated differential expression**—suggesting these genes are downstream targets of the same oncogenic driver or transcription factor.

5 Discussion and Clinical Interpretation

5.1 Summary of Statistical Findings

This study aimed to characterize the transcriptomic differences between paired tumour and normal tissues. After quality control (removal of `Sample003` and `Sample018`), the analysis proceeded with 28 samples. As summarized in Table 1, the Negative Binomial model identified a robust signature of malignancy.

Table 1: Summary of Differential Expression Results

Metric	**Count**
Total Genes Analyzed	4,568
Significant Genes (Raw $p < 0.05$)	3,266
Significant Genes (Bonferroni Adj $p < 0.05$)	**1,604**
Genes with Convergence Warnings	492
Significant Genes with Warnings (Bonf Adj $p < 0.05$)	383

5.2 Clinical Implications

Diagnostic Biomarkers and Therapeutic Targets

The 1,604 genes identified as significantly differentially expressed represent potential biomarkers. The strict multiple testing correction increases confidence in any one gene; Bonferroni minimises the chance of any false positives. Genes with high "fold-changes" (large coefficients in the NB model) are prime candidates for diagnostic panels, as their expression levels reliably distinguish tumour from normal tissue. Furthermore, genes that are significantly upregulated in tumours could serve as therapeutic targets for inhibition.

Tumour Heterogeneity and Personalized Medicine

The Principal Component Analysis (PC1 vs PC2) provided critical clinical insight. While PC1 (30.8% variance) clearly distinguished disease state, the vertical dispersion of tumour

samples along PC2 highlights **inter-patient heterogeneity**. This implies that cancer is not a monolithic disease; while all patients share a common "malignant signature," individual tumors are molecularly distinct. This supports the paradigm of **Precision Medicine**: a "one-size-fits-all" treatment strategy may fail for patients clustering in different regions of the PCA plot. Future studies should investigate if these clusters correlate with clinical outcomes (e.g., survival, drug resistance).

The Cost of Paired Analysis in Small Samples

Our comparison of linear models revealed that controlling for patient identity (Paired Design) did not improve statistical power in this dataset. Despite the theoretical advantages of matched designs, the inclusion of 13 patient indicator variables in a cohort of only 28 samples substantially reduced residual degrees of freedom, inflating standard errors and yielding less significant results than the simpler unpaired model.

This finding does not invalidate paired designs in general. In larger cohorts, the reduction in unexplained variance achieved by blocking on patient identity would be expected to outweigh the degrees-of-freedom penalty. However, our results illustrate a critical consideration for clinical trial design: the benefits of matched sampling can only be realised when sample sizes are sufficient to accommodate the additional model complexity. For small studies, simpler unpaired analyses may retain greater statistical power despite not controlling for inter-individual variability.

5.3 Limitations and Future Work

- **Sample Size:** With only 14 effective pairs (n=28), the study is underpowered to detect subtle effects. The exclusion of `Sample003` was necessary for validity but further reduced the cohort size. Notably, the small sample size prevented the paired linear model from outperforming the unpaired model, as the degrees-of-freedom cost of estimating patient-level parameters exceeded the variance reduction benefit.

- **Model Instability:** Approximately 10% of genes generated convergence warnings, likely due to data sparsity (zero-inflation). These genes were flagged rather than removed, but their p-values must be validated using orthogonal methods (e.g., qPCR). In larger cohorts, the fit of the NB model should also be compared with

alternative non-parametric models (goodness-of-fit testing) to assess FDR control (Robinson and Oshlack 2010).

- **Exploratory Nature:** The co-expression analysis is hypothesis-generating. The inferred "rewired networks" require functional validation (e.g., Gene Ontology enrichment) to define the specific biological pathways involved.

- **Normalization Limitations:** The strong correlation between PC1 and library size ($r = -0.85$, $p < 0.001$) suggests that CPM normalization incompletely removed technical variation. Future analyses should consider more robust normalization methods such as TMM or quantile normalization.

5.4 Conclusion

This report demonstrates that probabilistic modelling (NB-GLM), combined with rigorous normalization and multiple testing correction, can effectively isolate cancer-specific gene expression signatures. The analysis also revealed important methodological insights: paired study designs, while theoretically superior, require adequate sample sizes to realise their statistical advantages. The findings highlight the dual importance of identifying universal cancer biomarkers while acknowledging the individual heterogeneity that necessitates personalized therapeutic approaches.

Declaration of Generative AI Usage

In accordance with coursework guidelines, Generative AI tools (Google AI Studio, Perplexity, Claude) were used for the following permitted purposes:

- **Code Generation:** AI was used to generate and debug R code syntax, which was subsequently reviewed, modified, and fully commented by the author.
- **Conceptual Clarification:** AI was used as a learning aid to clarify statistical concepts (e.g., degrees of freedom in paired designs), enabling the author to independently interpret results.
- **Proofreading:** AI assisted with grammar checking and structural organisation of the report.

Confirmation: All statistical interpretations, biological conclusions, and written analysis are the original work of the author.

Appendix: Example AI Prompt and Output

Prompt: "Write an R loop to fit two linear models for each gene—one with tissue only and one with tissue plus patient ID—and extract the p-values for the tissue coefficient."

Generated Code:

```
for (g in 1:G) {
  y_g <- Y_log_clean[g,]

  fit1 <- lm(y_g ~ x)
  p_lin1[g] <- summary(fit1)$coefficients["xTumour", "Pr(>|t|)"]

  fit2 <- lm(y_g ~ x + z)
  p_lin2[g] <- summary(fit2)$coefficients["xTumour", "Pr(>|t|)"]
}
```

This code was integrated into the analysis pipeline by the author.

References

Clyte.tech (2023). *Bonferroni vs FDR: Multiple Hypothesis Testing in Biomedical Data*. URL: https://www.clyte.tech/post/bonferroni-vs-fdr-multiple-hypothesis-testing-in-biomedical-data (visited on 01/10/2026).

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Statomics (n.d.). *Sequencing Count Data*. URL: https://statomics.github.io/SGA/sequencing_countData.html (visited on 01/10/2026).